

Elastic contact forces and independent angular couples

Verified sphere–plane examples

Rigid Body Collisions research prototype

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The model is a synthetic constitutive hypothesis, not a calibrated rubber law or a new friction law. It verifies an independent collision couple, explicit elastic storage, yielding and conditional reversal. Frozen source: 9b0d2d80e20c. 4 of 10 cases satisfy the registered gates; 28 histories and 2 rejected attempts are retained. Numerical timings were collected concurrently with native friction research and cannot establish a general speed ranking.

A point-like wrench includes a couple

For a sphere of mass m , radius R and inertia $\mathbb{I} = \frac{2}{5}mR^2$, a fixed plane has normal $\hat{n}$ directed into the admissible half-space. The reference lever is $r = -R\hat{n}$; compression is $\delta = \max(0, R - \hat{n} \cdot x + d)$.

$$r\wedge = \begin{bmatrix} 0 & -r_z & r_y \\ r_z & 0 & -r_x \\ -r_y & r_x & 0 \end{bmatrix}, \quad v_c = v + (r\wedge)^\top \omega.$$

$$\Delta p = \int F dt, \quad \Delta L = \int \tau_n \hat{n} dt, \quad m\Delta v = \Delta p + \int mg dt, \quad \mathbb{I}\Delta\omega = (r\wedge)\Delta p + \Delta L.$$

The normal-axis spin change requires the independent couple ΔL : the offset point force has zero torque component along $\hat{n}$. A point-like wrench is a representation of effective contact mechanics, not a claim that a mathematical point traction carries a moment.

Elastic history, shared yield and energy

Let q be the two-component tangential displacement, θ the relative twist angle and a an effective material contact length. Define $h = (q_1, q_2, a\theta)^\top$, $K = \text{diag}(k_t, k_t, k_\theta/a^2)$ and $f = (\delta/R)^p$, with $p = 2$ for the default compression-weighted contact.

$$U = \frac{1}{2}k_n\delta^2 + \frac{1}{2}fh^\top Kh, \quad U_h = \frac{1}{2}fh^\top Kh, \quad F_n = k_n\delta + c_n \max(0, \dot{\delta}) + pU_h/\delta.$$

$$F_t = -fk_t q, \quad \tau_n = -fk_\theta \theta, \quad \sqrt{\|F_t\|^2 + (\tau_n/a)^2} \leq \mu k_n \delta \leq \mu F_n.$$

The last bound shares the normal-load budget between sliding and twisting. It is a phenomenological ellipsoidal capacity approximation, not an exact patch traction result. Using the elastic base normal load makes the budget conservative even when history or damping adds normal force. One coefficient μ covers sticking and sliding; this prototype does not yet fit distinct static and dynamic coefficients. Damping acts only during compression.

Let T contain orthonormal plane tangents and $u = (T^\top v_c, a\hat{n} \cdot \omega)^\top$. During elastic stick, $\dot{h} = u$. At the yield boundary, associated plastic flow has $\dot{h} = u - \lambda s$, with $s = fKh/\|fKh\|$ and $\lambda \geq 0$ chosen to prevent outward motion of the yield function. Consequently

$$\dot{D} = c_n \max(0, \dot{\delta})\dot{\delta} + \lambda\|fKh\| \geq 0, \quad \frac{d}{dt} \left(\frac{1}{2}m\|v\|^2 + \frac{1}{2}\mathbb{I}\|\omega\|^2 - mg \cdot x + U + D \right) = 0.$$

The default store vanishes smoothly at separation. The optional constant-stiffness benchmark uses $p = 0$ and explicitly records any residual history store as separation loss; matched oscillator tests return it to zero. No elastic state is silently deleted. Adaptive integration rejects an exhausted RHS evaluation budget.

Independent exact oscillators

For the linear benchmark $p = 0$, no damping and no yielding, choose

$$\omega_c = \sqrt{k_n/m}, \quad t_c = \pi/\omega_c, \quad k_\theta = \mathbb{I}k_n/m, \quad k_t = \frac{\mathbb{I}k_n}{\mathbb{I} + mR^2}.$$

The normal and twist modes then share a half period: $v_n^- = -V \rightarrow v_n^+ = V$, $\Omega_n^+ = -\Omega_n^-$. Throughout contact, $|\tau_n|/F_n = \mathbb{I}|\Omega_n^-|/(mV)$; the no-yield requirement is therefore $\mu a \geq \mathbb{I}|\Omega_n^-|/(mV)$, not merely an endpoint impulse inequality.

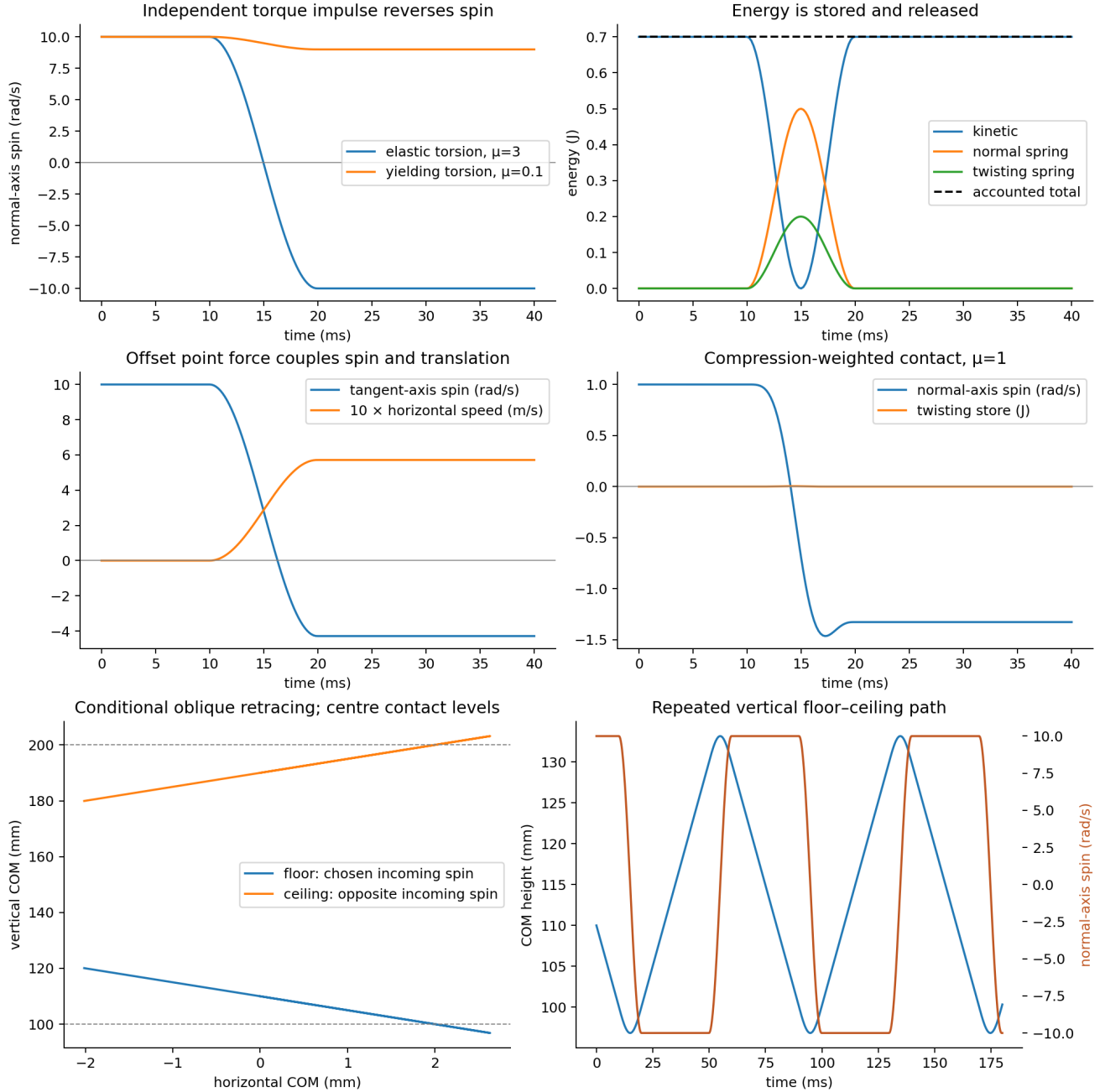
For a solid sphere dropped with zero horizontal COM speed and tangent-axis spin Ω , the same elastic tangential oscillator gives

$$\Omega_t^+ = -\frac{3}{7}\Omega_t^-, \quad v_t^+ = \frac{4}{7}R\Omega_t^-.$$

With $v_t^- = v_0$ and appropriately oriented $\Omega_t^- = -v_0/(\alpha R)$ at the floor, $\alpha = \mathbb{I}/(mR^2) = 2/5$, both horizontal velocity and spin reverse. A ceiling needs the opposite incoming spin. Arbitrary rubber-ball throws do not necessarily retrace. A vertical floor–ceiling path with pure normal-axis spin can repeat these exact elastic reversals.

Evidence and limitations

Case	Gates	Fine outgoing spin (rad/s)	Energy error (J)
elastic normal axis spin	no	0, 0, -10	2.2e-14
elastic tangent axis spin	yes	0, -4.2857, 0	6.7e-15
oblique floor retrace	yes	0, 5, 0	5.7e-15
oblique ceiling retrace	yes	0, -5, 0	2.8e-14
vertical floor ceiling	no	0, 0, -10	3.8e-14
low friction no spin reversal	no	0, 0, 9	1.4e-14
compression weighted reversal	no	0, 0, -1.3268	5.6e-15
slow same material	yes	0, 0, -0.1	6e-15
rapid same material	no	0, 0, -1000	6.5e-11
weighted high spin low friction budget	no	rejected	–



The weighted contact can transfer energy between normal and rotational channels, so a single-channel restitution may exceed one while total mechanics remain passive. Low friction can dissipate enough elastic input to prevent reversal. The same-material slow and rapid examples use speeds 0.01 and 100 m/s, demonstrating numerical behavior of this ideal law, not real rubber at those rates. All traces, failures, frozen parameters, source and three-level refinements are archived. This is a sphere/plane prototype; arbitrary-shape simultaneous contact needs separate engine validation. The accompanying literature review distinguishes known soft-finger mechanics from any unproven publication novelty.